

Mike Spencer

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EDUCATION

UNIVERSITY OF EDINBURGH

PHD IN SNOW HYDROLOGY
June 2016

LANCASTER UNIVERSITY

BSC IN ENVIRONMENTAL SCIENCE
June 2004

LINKS

Academic Blog: scottishsnow
Mastodon: [@MikeRSpencer](https://mikeerspencer@mastodon.social)
OrCID: [0000-0003-1472-2082](https://orcid.org/0000-0003-1472-2082)
LinkedIn: [mikerspencer](https://www.linkedin.com/in/mikerspencer)
GitHub: [mikerspencer](https://github.com/mikerspencer)
StackExchange: [mikerspencer](https://stackoverflow.com/users/mikerspencer)

MEMBERSHIP

Royal Statistical Society fellow
Society of Research Software Engineers
EdinbR user group convener
Software and data carpentry trainer

SKILLS

COMPUTING

Operating systems:
Linux (Fedora & Debian based)
Programming:
R • SQL • Git • Bash • $\LaTeX$ • Python •
HTML
Spatial:
QGIS • GRASS • ArcGIS • MapInfo Pro
Other:
Microsoft/Libre Office • Desktop
publishing, e.g. Scribus • Containers •
Knowledge management

DATA

Types:
SQL/databases • Arrow/parquet/Hive •
Time series • Numeric • Categorical (e.g.
social) • Surveys • Grids (e.g. satellite):
NetCDF, rasters, hdf, images •
Qualitative/text
Abilities:
Problem definition • Requirements •
Mining • Extraction •
Wrangling/munging • Management •
Quality assurance • Reproducibility •
Exploration • Analysis • Modelling:
statistical/physically based/machine
learning • Visualisation • Forecasting •
Interpretation • Insight • Storytelling

ABOUT

OVERVIEW

Mike is a research data science leader who enjoys diverse challenges. He has worked in the public, private, academic and third sectors; he has a tested record of quick learning and adaptability. Mike's experience across many domains helps him understand customer needs and find suitable approaches to solve difficult challenges.

Mike is both a leader and practitioner of data science and engineering, and is equally at home across these spaces. He began his career in the hydrology and environmental sectors before broadening his experience in agriculture, economics, finance and social care. Mike helps the wider community with expert data advice on scaling workflows and visualisation, and runs training sessions on databases, programming, GIS and other tools.

INTERESTS

- Using data science and engineering to improve workflows.
- Understanding spatial dimensions of business problems.
- Data storytelling and audience engagement.
- Promotion of open source software and data literacy.

SKILLS PORTFOLIO

LEADERSHIP AND DEVELOPMENT

- Mike has built and run two successful data science and engineering teams, comprising people from a range of backgrounds and who work across a number of domains and tools (e.g. R, Python, Arrow, dashboards, finance, climate, energy, health, and more).
- He is a hands-on technical leader who often builds proof-of-concept examples to test ideas and support change by demonstrating potential approaches to the teams he works with.
- He is comfortable leading and contributing to projects on data pipelines, architecture, software, deployment, research, modelling, dashboards, visualisation, documentation and more.
- Mike has run a number of high profile skill development events including: a [data visualisation workshop](#) on behalf of the Scottish Environment, Food and Agriculture Research Institutes and for DataFest 2018 a [network analysis using R](#) event.

POLICY IMPACT

- Mike was the lead data scientist on a proof of concept with a local authority to help them support people in the cost of living crisis. He developed a viable dataset and tool, while leading engagement with the local authority. This work is now in production and won the 2023 social impact category in the Scottish Financial Technology Awards.
- In 2018 Mike analysed the social and economic benefits of hill farming. This combined open and national datasets. For example environmental, like land-cover and elevation, along with population and agricultural census. [Writing code in R and GRASS GIS](#) he used machine learning to classify landuse.
- To improve resilience and efficiency in the beef industry, Mike used social network analysis to interpret millions of location-based interactions across Scotland. This work was used by the Scottish Government to develop a replacement for the EU common agricultural policy and [explores farmer interactions, economic value-chains and efficiency metrics](#).

OUTPUT

ARTICLES (SELECTED)

Pohle, I., Helliwell, R., Aube, C., Gibbs, S., Spencer, M. & Spezia, L., *Citizen science evidence from the past century shows that Scottish rivers are warming*, Science of The Total Environment, **2019**, 659, p53-65.

Ge J., Polhill J.G., Matthews K.B., Miller D.G. & Spencer M., *Not one Brexit: How local context and social processes influence policy analysis*, PLOS ONE, **2018**, 13(12): e0208451.

Spencer, M., *Snow Survey of Great Britain: transcribed data for Scotland, 1945 to 2007*, NERC Environmental Information Data Centre, **2016**.

Spencer, M. & Essery, R., *Scottish snow cover dependence on the North Atlantic Oscillation index*, Hydrology Research, **2016**.

Spencer, M., Essery, R., Chambers, L. & Hogg, S., *The Historical Snow Survey of Great Britain: Digitised Data for Scotland*, Scottish Geographical Journal, **2014**, 130, p252-265.

TALKS (SELECTED)

Quarto for Python and R, Research Software Practices in the Social Sciences, Software Sustainability Institute, **Mar 2025**

Opening Finance for Good, Bank of England invited research seminar, **Nov 2023**

*Beyond the *IMD - filling the data gap*, FOSS4GUK, Glasgow University, **Oct 2023**

Data science: open understanding, Lancaster University DSNE, **Dec 2021**

Mapping access to services to estimate community resilience, Edina geoforum, **Mar 2020**

Cartography with R and QGIS, which to choose?, FOSS4GUK, **Sep 2019**

Predicting snowmelt with R, Royal Statistical Society, **Apr 2017**

Snow, R and keeping people safe, EARL, **Sep 2016**

Scottish snow cover and the North Atlantic Oscillation, British Hydrological Society (BHS) symposium, **Sep 2014**

POSTERS (SELECTED)

Beef efficiency - finishing systems., RESAS end of year, **Mar 2020**

Snowmelt frequency in a mountainous, temperate, maritime environment, International conference on alpine meteorology, **Sep 2015**

Snow survey of Great Britain: digitised data for Scotland, BHS symposium, **Sep 2014**

ZENODO (FULL)

Browse all Mike's uploads to the [open science platform](#). Including code, datasets, posters, presentations and graphics.

EXPERIENCE

MIKE R SPENCER | DATA CONSULTANT

Aug 2020 - ongoing | Midlothian, Scotland

- Data science associate with Data Orchard.
- Review and QA of [Data Orchard's data maturity assessment tool](#).
- Support the UK Government Global Monitoring and Evaluation programme.
- Designed data architecture and wrote extract, transform & load pipelines for £1B Joint Funds Unit interactive dashboards.

SMART DATA FOUNDRY | PRINCIPAL DATA SCIENTIST

Sep 2021 - Apr 2025 | Edinburgh, Scotland

- Built and lead a team of 7 research data scientists and engineers.
- Social impact winner in the Scottish Financial Technology Awards.
- Conference and seminar speaker.
- Client facing research and insights engagement.
- Technical lead for software stack, data engineering and data science projects.

VARIOUS | LEAD DATA ROLES

Nov 2020 - Mar 2026 | Remote, UK

- Scottish Prison Service: Improved data engineering and reporting through training and demonstration projects.
- NSPCC: lead team working on AI summarisation, NLP and propensity modelling.
- Scottish Social Services Council: tech lead for workforce intelligence team.

AGRIMETRICS | EXECUTIVE LEADERSHIP TEAM - DATA SPECIALIST

Aug 2018 - Oct 2020 | Rothamsted, England

- Provide data oversight for the UK's leading agri-food data science centre.
- Convened sessions between Agrimetrix, SRUC and Scottish Government on crop mapping from remotely sensed data.
- Represented SRUC in a collaboration across multiple agritech organisations.

SRUC | RESEARCH DATA SCIENTIST

Jan 2017 - Oct 2020 | Edinburgh, Scotland

- Expert data advice to 45 social researchers and students.
- Linking and analysis of: animal movements, networks, surveys, spatial data.
- Developed and ran internal and external data skills training series.
- Producing high-quality visualisations, e.g. web-apps, leaflets, and posters.

HOSTELLING SCOTLAND | DIRECTOR & BOARD ADVISOR

Jun 2016 - Jun 2019 | Edinburgh, Scotland

- Advertised, interviewed and appointed a new CEO.
- Process development and governance for board member recruitment.

UNIVERSITY OF EDINBURGH | PHD

Sep 2011 - Jun 2016 | Edinburgh, Scotland

- Stochastic time series snow modelling and model build.
- Extreme value statistics of snow cover and melt.
- Analysis of remote sensing veracity using machine learning.
- Data mining of historic Snow Survey of Great Britain dataset.
- Extensive fieldwork in the Arctic and Scotland.

VARIOUS | HYDROLOGIST

2005 - 2011 | England, Wales & Scotland

- Private sector: Halcrow Group Ltd and Arcadis GMI.
- Public sector: Environment Agency Wales.
- Flood risk, hydrological design, contaminated land, drought, etc..

PERSONAL

Mike has lived on the outskirts of Edinburgh since Autumn 2008. He balances his work commitments with family life. When he's not at a computer Mike can be found on a bike, mountain, or gigging as a trombonist.